Priyanshu Dasgupta

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Education

Princeton University — Princeton, NJ
A.B. Physics; Minors: Materials Science & Eng., Optimization & Quant. Decision Science — *Sep. 2026 – May 2030*

- **Coursework**: Classical to Quantum Mechanics (PHY 207), Single Variable Analysis (MAT 215), Statistics (ORF 245)
- **Dual Enrollment** (2025–26): PHY 105 Mechanics (A), PHY 106 Electromagnetism (A), Special Relativity (A+)
- **Activities**: Electric Speedboating (autonomy), HackPrinceton, Princeton University Energy Association, Tiger Investments Group (Education Sector), Tarana (a cappella), Physics Dept. UCA (PHY 105) & Web Dev

Montgomery High School — Skillman, NJ
GPA: 99.2/100 | SAT: 1550 | ACT: 35 | PSAT: 1500 — *Sep. 2022 – Jun. 2026*

- **Coursework**: AP Calc BC, Physics C, Chem, CS A; **Columbia SHP**: Plasma Physics, Astronomy, Rel. Astrophysics
- **Activities**: NHS; Science Board co-founder; Science Expo Board; Science Bowl Varsity (PPPL); CJML; ARML

Experience

Machine Learning Developer — 2026 – Present
Health-Tech Startup

- Developing **wearable tech** tracking **blood pressure**, **SpO_2** & other heart stats from in-house built hardware
- Designed **U-Net + LSTM** models in **Python** on device data, benchmarked vs. **EEG/ECG**, to detect heart disease

IPT Team Experimental Physicist + Data Analyst — Sep. 2026 – Present
Authentic Research Partners – Team USA, International Physicists' Tournament

- Recruited to represent **Team USA**; one of 5 members, in charge of **modeling simulations** for the team's papers

Research

Student Researcher — Jun. 2024 – Aug. 2025
Mikhailova ETOILES Lab & Fleischer Imaging Lab, Princeton University (via LLP) — *Princeton, NJ*

- Imaged an **argon current-carrying plasma double layer** with **Schlieren imaging** & **two-color interferometry**
- Built Schlieren airflow algorithms in **MATLAB/Python** for **3D** aircraft-turbulence models; color-dependent **BOS**
- Presented at the **APS Plasma Physics Conference**; first author & presenter, **nonlinear optics** poster, **APS 66th DPP**

Quarknet Particle Physics Program — Jul. 2025 – Aug. 2025
Rutgers University–New Brunswick — *New Brunswick, NJ*

- Analyzed **MINERvA neutrino** & **Fermilab** data; muon scintillators; **Qiskit**; talk to **100+** on **Grover's Algorithm**

Student Research Assistant — Mar. 2024 – Sep. 2024
Nokia Bell Labs (under Dr. Yury Deshko) — *Murray Hill, NJ*

- Studied **Zero Phonon Line** in Xe^+-implanted CVD diamond; modeled **V-Xe-V** centers in **Python/MATLAB**

Leadership & Activities

Science Olympiad | *A-Team Captain (MHS); Executive Board (Princeton SO)* — 2022 – Present

- **4th, 10th, 7th at Nationals** ('24–'26); 2 fundraisers (**$31K+ each**); Princeton SO test-quality lead & test writer

Kolkata Delicacies + MishtiSukh | *Co-Founder*; **30+ monthly clients**, ∼**$10K/yr** to Kolkata families — 2022 – Present

ELL Program @ Princeton | *Founder*; lead English teacher for **40+ refugees**, 79.5 hrs/yr — 2022 – 2026

TEAMS | *VP & Captain*; **4× Nationals qualifier**; 3rd & 2nd at States ('25, '26) — 2022 – 2026

Music | *Regional Director*, BackToBach; lead guitar/vocals, *Ghore Ferar Gaan*; Rider Univ. Guitar Ensemble — 2017 – Present

Honors & Awards

USAPhO Gold Medal (2026) | **USACO** Gold (2025) | MHS Math & Physics Dept. Awards
Provisional Patent (expired): STREAMS, a programmable modular wind-tunnel fan array
North NJ JSHS (2025): 3rd Place Poster Presenter overall, 1st Place Physical Sciences
Science Olympiad: 49 medals; 33 top-5 at MIT, Cornell, Princeton, Columbia & UPenn; 7 golds @ States, 2 silvers @ Nats

Skills

Programming & Libraries: Python, MATLAB, Java, Excel; PyTorch, scikit-learn, Matplotlib, Qiskit; Django, Flask, React
Methods: Deep learning (U-Net, LSTM), modeling & simulation, experimental design, Schlieren & BOS imaging, interferometry
Languages: English, Bengali & Hindi (native), Spanish (professional), Russian (elementary)